Assignment No 6

Aim: Implement Linear Regression Algorithm

Theory:

Linear Regression is a **supervised machine learning algorithm** used to predict a continuous numerical value based on one or more independent variables. It establishes a mathematical relationship between the input variables and the output variable.

For example, Linear Regression can be used to predict a student's **marks based on study hours**, predict **house prices based on area**, or predict **sales based on advertising expenditure**.

The main objective of Linear Regression is to find the **best-fit line** that represents the relationship between the independent variable and dependent variable.

Basic Equation

For **simple linear regression**, where there is only one independent variable, the equation is:

$$\hat{Y} = b_0 + b_1 X$$

Where:

- $\hat{\mathbf{Y}}$ = Predicted value of the dependent variable
- **X** = Independent variable
- $\mathbf{b_0}$ = Intercept of the regression line
- $\mathbf{b_1}$ = Slope or regression coefficient

The **slope ($\mathbf{b_1}$)** represents how much the dependent variable is expected to change when the independent variable increases by one unit.

For example, if the regression equation is:

$$Marks = 30 + 5(Hours)$$

then the intercept is **30** and the slope is **5**. This means that for every additional hour of study, the predicted marks increase by approximately 5 marks.

Types of Linear Regression

1. **Simple Linear Regression:**
   Uses one independent variable to predict one dependent variable.

Example:
Study Hours → Marks

2. **Multiple Linear Regression:**
   Uses two or more independent variables to predict one dependent variable.

Example:
Study Hours + Attendance + Assignment Score → Marks

Working of Linear Regression

The main steps involved are:

1. Collect the dataset.
2. Identify the independent and dependent variables.
3. Preprocess the data if required.
4. Divide the dataset into training and testing sets.
5. Train the Linear Regression model using the training data.
6. Find the best-fit regression line.
7. Use the trained model to predict values for the test data.
8. Evaluate the performance of the model.

Least Squares Method

Linear Regression commonly uses the **least squares method** to find the best-fitting line. It minimizes the difference between the actual values and the predicted values.

The difference between an actual value and its predicted value is called a **residual or error**.

The model attempts to minimize the **sum of squared errors**:

$$SSE = \sum (Y - \hat{Y})^2$$

A good regression model generally has smaller prediction errors.

Evaluation Metrics

Several metrics can be used to evaluate a Linear Regression model:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between actual and predicted values.
- **Mean Squared Error (MSE):** Calculates the average squared prediction error.
- **Root Mean Squared Error (RMSE):** Square root of MSE.

- **R^2 Score:** Indicates how well the independent variables explain the variation in the dependent variable.

An **R^2 score closer to 1** generally indicates that the model explains a large proportion of the variation in the target variable.

Advantages

- Simple and easy to understand.
- Easy to implement.
- Computationally efficient.
- Useful for predicting continuous numerical values.
- The relationship between variables can be easily visualized.

Limitations

- Assumes a linear relationship between variables.
- Sensitive to outliers.
- Performance can be affected by highly correlated independent variables.
- Not suitable for problems where the relationship is strongly non-linear.

Applications

Linear Regression is commonly used for:

- House price prediction.
- Sales forecasting.
- Student marks prediction.
- Salary prediction.
- Demand forecasting.
- Business and financial analysis.
- Temperature prediction.

Conclusion

Linear Regression is an important supervised learning algorithm used to predict continuous numerical values. It establishes a relationship between independent and dependent variables by finding the best-fit line and can be evaluated using metrics such as **MAE, MSE, RMSE, and R^2 score**.

Code: import pandas as pd

```
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt

housing = fetch_california_housing()
X = housing.data
y = housing.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

y_pred = lr_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Coefficients:", lr_model.coef_)
print("Intercept:", lr_model.intercept_)
print("Mean Squared Error:", mse)
print("R² Score:", r2)
```

```
plt.scatter(y_test, y_pred, alpha=0.5)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted Housing Prices")
plt.show()
```

